

PETRUN', Russian Federation

WMO: 233240

Lat: **66.4711N**Lon: **60.7544E**Elev: **62**StdP: **100.58**Time Zone: **3.00 (E03)**Period: **09-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -40.4	(c) -37.5	(d) -42.9	(e) 0.1	(f) -40.1	(g) -40.2	(h) 0.1	(i) -37.4	(j) 9.0	(k) -11.9	(l) 8.2	(m) -10.3	(n) 2.0	(o) 20	(p) 0.754

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 10.7	(c) 27.4	(d) 18.4	(e) 25.1	(f) 17.3	(g) 22.5	(h) 16.0	(i) 19.3	(j) 25.9	(k) 18.1	(l) 23.7	(m) 16.8	(n) 21.4	(o) 2.8	(p) 140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.9	12.2	21.3	15.7	11.2	20.6	14.4	10.3	19.8	55.3	25.6	51.5	23.8	47.3	21.4	22.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7.7	7.0	6.2	-40.9	29.4	3.9	2.2	-43.7	31.0	-46.0	32.2	-48.2	33.5	-51.1	35.0
			-40.8	20.8	3.6	1.3	-43.3	21.7	-45.5	22.5	-47.5	23.3	-50.1	24.2
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	-2.8	-20.1	-17.3	-11.2	-4.4	1.8	10.6	14.9	10.8	6.3	-1.4	-10.7	-14.3
	DBStd	13.41	9.52	9.52	8.20	5.81	5.84	5.29	4.77	4.06	3.82	5.24	8.51	10.07
	HDD10.0	5004	933	766	656	433	266	55	8	39	123	353	621	753
	HDD18.3	7757	1192	999	914	683	512	237	131	235	361	612	871	1012
	CDD10.0	322	0	0	0	0	12	74	160	64	12	0	0	0
	CDD18.3	32	0	0	0	0	0	6	24	2	0	0	0	0
	CDH23.3	389	0	0	0	0	3	68	296	22	0	0	0	0
	CDH26.7	82	0	0	0	0	0	9	71	2	0	0	0	0
Wind	WSAvg	3.2	2.9	3.1	3.3	3.5	3.4	3.3	2.8	2.9	3.0	3.2	3.3	3.3
Precipitation	PrecAvg	488	29	25	25	25	39	42	58	66	58	51	39	33
	PrecMax	688	54	43	55	59	71	86	121	128	112	104	67	59
	PrecMin	346	11	2	5	11	12	16	11	17	0	21	18	10
	PrecStd	89	11	10	11	12	13	18	31	29	26	18	13	14
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-0.1	0.4	3.7	9.4	23.5	28.1	30.3	26.4	20.5	8.2	2.5	0.9
		MCWB	-0.9	-0.6	0.7	4.8	13.8	18.0	19.2	18.9	15.0	6.5	2.2	0.3
	2%	DB	-2.4	-1.1	1.9	6.7	19.8	25.5	28.8	22.7	16.6	6.7	0.9	0.0
		MCWB	-3.1	-2.2	0.0	3.0	12.6	16.7	19.1	16.9	13.3	5.5	0.3	-0.6
	5%	DB	-5.0	-2.6	0.8	4.6	15.3	23.0	26.8	20.4	14.2	5.5	0.3	-1.0
		MCWB	-5.5	-3.4	-0.5	2.0	10.1	15.8	18.3	15.4	11.5	4.5	-0.2	-1.8
	10%	DB	-7.4	-4.7	-0.5	3.1	11.1	20.3	24.6	18.1	12.1	4.4	-0.8	-2.4
		MCWB	-8.0	-5.6	-1.5	1.2	7.1	14.5	17.2	14.3	10.1	3.6	-1.3	-3.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-0.7	-0.2	1.2	4.9	14.7	18.9	20.6	20.1	15.5	7.3	2.2	0.5
		MCDB	0.0	0.4	2.9	8.8	21.8	25.1	26.7	23.7	18.9	8.1	2.3	0.7
	2%	WB	-3.1	-2.0	0.4	3.5	12.6	17.7	19.8	17.7	13.9	5.7	0.6	-0.4
		MCDB	-2.4	-1.2	1.5	6.0	18.9	23.7	27.0	21.2	16.1	6.3	0.8	-0.1
	5%	WB	-5.6	-3.5	-0.3	2.4	10.0	16.2	19.0	16.2	11.6	4.6	-0.1	-1.7
		MCDB	-5.0	-2.6	0.8	4.0	14.9	21.7	25.9	19.3	13.4	5.3	0.3	-1.1
	10%	WB	-8.1	-5.5	-1.5	1.5	7.4	14.8	17.9	14.7	10.3	3.7	-1.3	-3.1
		MCDB	-7.5	-4.6	-0.4	2.9	10.9	20.2	23.5	17.4	11.8	4.3	-0.9	-2.4
Mean Daily Temperature Range	5% DB	MDBR	7.8	7.8	9.3	8.9	8.2	9.2	10.7	8.3	6.6	4.3	6.9	6.4
		MCDBR	6.6	6.1	6.2	8.8	14.6	13.2	14.3	12.6	9.8	4.5	4.1	4.9
		MCWBR	6.4	5.7	5.1	6.0	8.3	6.5	6.4	7.1	6.3	3.6	4.3	4.9
	5% WB	MCDBR	6.3	6.0	5.1	7.4	14.2	12.3	13.3	10.7	8.6	4.3	4.0	4.9
		MCWBR	6.2	5.7	4.5	5.3	8.3	6.8	6.6	6.9	5.8	3.6	4.2	4.9
Clear-Sky Solar Irradiance	taub	0.212	0.268	0.308	0.336	0.349	0.359	0.391	0.367	0.325	0.281	0.207	0.219	
	taud	1.886	2.019	2.056	2.084	2.248	2.438	2.352	2.396	2.471	2.244	1.923	1.899	
	Ebn at Noon	173	551	725	809	846	850	802	784	740	553	167	10	
	Edh at Noon	17	58	94	120	113	97	103	89	65	46	15	3	
All-Sky Solar Radiation	RadAvg	0.06	0.59	2.08	3.89	4.90	4.98	4.95	3.08	1.77	0.57	0.13	0.01	
	RadStd	0.01	0.09	0.24	0.32	0.53	0.46	0.72	0.34	0.19	0.09	0.02	0.00	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (3 neighbors)		+1.22	N/A	N/A	N/A	N/A	N/A	-342	-405	N/A	N/A

Nomenclature: See separate page